Actor–Critic Methods

Mattiahs Bubich

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1 Main Goal

Combine:

- A **policy** (the actor) updated via policy gradients,
- A **value function** (the critic) used both as a baseline and to assess actions,

to achieve **lower variance**, **online learning**, and **faster convergence**.

2 From REINFORCE with Baseline to Actor–Critic

- In REINFORCE with baseline:
  - The value function $\hat{v}(S_t, w)$ serves only as a baseline.
  - It is computed *before* selecting the action A_t.
- In Actor–Critic:
  - After taking action A_t, observe R_{t+1} and S_{t+1}.
  - Use the **one-step return** to directly assess the action taken.

3 Key Concept: One-Step Return

The one-step return is defined as:

$$G_{t:t+1} = R_{t+1} + \gamma \hat{v}(S_{t+1}, w)$$

This provides a quick, biased but lower-variance estimate of the true return.

4 Why Prefer One-Step Return

- Allows **incremental updates** at every step.
- Enables **online learning**.
- Greatly **reduces variance**, despite introducing some bias.

This parallels the advantage of TD(0) over Monte Carlo methods in value prediction.

5 Actor–Critic Updates: Full Derivation

6 TD Error

The temporal difference (TD) error is:

$$\delta_t = R_{t+1} + \gamma \hat{v}(S_{t+1}, w) - \hat{v}(S_t, w)$$

7 Actor Update

Policy parameters θ are updated as:

$$\theta_{t+1} = \theta_t + \alpha_\theta \delta_t \nabla \log \pi(A_t | S_t, \theta_t)$$

8 Critic Update

Value function parameters w are updated via semi-gradient TD(0):

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla \hat{v}(S_t, w_t)$$

9 Interpretation of the Updates

- The **critic** evaluates actions based on the TD error.
- The **actor** adjusts the policy based on how good the taken action was relative to expectation.

10 Full Pseudocode: One-Step Actor–Critic (Episodic)

Inputs:

- Policy $\pi(a|s, \theta)$,
- Value function $\hat{v}(s, w)$.

Algorithm:

1. Initialize θ, w.
2. For each episode:
    - Initialize S.
    - Loop while S is not terminal:
        - Sample $A \sim \pi(\cdot|S, \theta)$.
        - Take action A, observe R and S'.

- If S' is terminal, set $\hat{v}(S', w) = 0$.
- Compute TD error:

$$\delta = R + \gamma \hat{v}(S', w) - \hat{v}(S, w)$$

- Update critic:

$$w \leftarrow w + \alpha_w \delta \nabla \hat{v}(S, w)$$

- Update actor:

$$\theta \leftarrow \theta + \alpha_\theta \delta \nabla \log \pi(A|S, \theta)$$

- Set $S \leftarrow S'$.

11 Extensions: n-Step and λ-Return Actor–Critic

Replace $G_{t:t+1}$ by:

- n-step return $G_{t:t+n}$,
- or λ-return G_t^λ.

Eligibility traces:

- Maintain separate traces z_θ, z_w for actor and critic,
- Traces decay according to λ_θ and λ_w.

12 Full Pseudocode: Actor–Critic with Eligibility Traces (Episodic)

Inputs:

- Policy $\pi(a|s, \theta)$,
- Value function $\hat{v}(s, w)$.

Parameters:

- Step sizes $\alpha_\theta > 0$, $\alpha_w > 0$,
- Trace-decay rates λ_θ, λ_w.

Algorithm:

1. Initialize θ, w, $z_\theta = 0$, $z_w = 0$.
2. For each episode:
   - Initialize S.

- Loop while S is not terminal:
  - Sample $A \sim \pi(\cdot|S, \theta)$.
  - Take action A, observe R and S'.
  - If S' is terminal, set $\hat{v}(S', w) = 0$.
  - Compute TD error:

$$\delta = R + \gamma \hat{v}(S', w) - \hat{v}(S, w)$$

  - Update eligibility traces:

$$z_w \leftarrow \gamma \lambda_w z_w + \nabla \hat{v}(S, w)$$

$$z_\theta \leftarrow \gamma \lambda_\theta z_\theta + \nabla \log \pi(A|S, \theta)$$

  - Update parameters:

$$w \leftarrow w + \alpha_w \delta z_w$$

$$\theta \leftarrow \theta + \alpha_\theta \delta z_\theta$$

  - Set $S \leftarrow S'$.

13 Comparison: REINFORCE vs Actor–Critic

Feature	REINFORCE	Actor–Critic
Return used	Full G_t	One-step or n-step $G_{t:t+n}$
Variance	High	Lower
Bias	None	Some bias
Learning mode	Episodic	Online
Critic usage	Only baseline	Full evaluation

14 Final Takeaways

- Actor–Critic methods combine the sample efficiency of TD learning with the flexibility of policy gradients.
- They allow for continuous learning without waiting for episode terminations.
- They are the foundation for modern algorithms like A2C, A3C, PPO, TRPO, DDPG, and SAC.